Discovering new classes at 2σ: the sigma-currency campaign

exps 146–151: CIFAR-10, CIFAR-100 (leakage-free) and Galaxy10 — summary and complete tables

P. C. Harris

September 7, 2026

The measurement

Every number in this document is one currency: $f^\star(2\sigma)$, the smallest fraction of a held-out class injected into a 5000-point corpus at which the *median expected significance* of a dataset-level two-sample test reaches 2σ against a 200-toy null (50 signal toys per fraction, add-one empirical tail, log-interpolated between grid fractions; lower is better). It is the discovery question asked the way an experiment would ask it: how rare can the new class be before this pipeline sees it at 2σ? Tests: mean min-Euclidean distance, mean min-Mahalanobis *(Maha.)*, 3-bandwidth MMD^2, the SparKer NP kernel ensemble, and — post-discovery only — two *anchor-aware* statistics that consume the anchors the discovery loop plants: the mean of $d_{\mathrm{seen}} - d_{\mathrm{disc}}$ *(Eucl. (anch.))* and SparKer with kernels seeded at the discovered anchors *(SparKer (anch.))*.

Coverage

CIFAR-10, leakage-free: all 8 objectives of the design cube plus 16 residual constructions, trained from random initialisation (the encoder never saw the held-out class), 4 holdout draws; discovery mode under three loop variants (distance pool with the 95th-percentile cut; distance and density-ratio pools with the paper's derived label-free cut, skip-on-refuse) and an n_{min} sweep. *Galaxy10*: 14 spaces (6 fine-tuned objectives, 2 transductive GCD arms, 6 residual constructions) on three pretrained backbones, 5 holdout draws each, pre-discovery; discovery mode (density-ratio pool, derived cut, n_{min}=5) on the three supervised arms and the three cross-draw residual children. *CIFAR-100* (same from-scratch lineage, b=0.01): all 8 objectives plus 16 residual constructions, 4 draws, pre-discovery. Every cell is single-seed; the campaign's variance decomposition (seed spread an eighth of draw spread) is why intervals are over draws.

Findings

1. **Neither the probe nor ranking AUC predicts $f^\star$.** The best-probed spaces are rarely the most detectable (CIFAR: probe 0.97 spaces sit mid-table; NPLM-dist. probes at 0.79 yet detects at 0.027), so representation quality for discovery has to be measured in this currency, not proxied.

2. **The best anchor-blind test is dataset-dependent, but always a density-ratio test.** SparKer leads on the compact CIFAR scratch heads (best or tied on 17/20 spaces, never censored); MMD leads on the Galaxy10 transfer heads (essentially never censored there, while SparKer and Mahalanobis are censored on 1–4 of 5 draws for half the rows).

3. **The residual construction earns a place, and its role scales with class count and with discovery.** Frozen, it is a stabiliser at 10 classes, and at 100 classes the concat sets the record (SupCon→res 0.018 vs. parent 0.023). Through the discovery loop, the *contrastive* child of SupCon becomes the best pipeline on 2 of 3 Galaxy10 backbones (0.013/0.015 vs. parents' 0.018/0.020) and rescues the one CIFAR cell where the parent's critic abstains entirely

(holdout 4: child engages at purity 0.2–0.4 and reaches 0.015). The NPLM child ties its parent throughout: it is the contrastive residual that recovers discarded novelty.

4. **Discovery pays only through the anchors it plants.** Anchor-blind tests *degrade* after the fine-tuning loop on both datasets; the anchor-aware statistics convert the same loop into the best pipelines measured. The loop beats the frozen space in 38/48 CIFAR (arm, draw, variant) cells and in all 9 Galaxy10 (backbone, arm) blocks.

5. **The pool decides, and the density-ratio pool with the derived cut supplies it.** Where the distance pool gives SupCon purity 0.000 (and pathological anchors), the np pool gives 0.3–0.9 on CIFAR and median purity 1.00 on Galaxy10 — and $f^\star$ follows. The scorer, not the cut, is what makes the pool pure; the cut sets its size and, when the estimated novelty is below n_{min}, *declines* — every remaining loss is such an abstention (daggers in the tables), and the one arm that declines everywhere (SupCon at holdout 4, all n_{min}) is exactly the arm whose pools are irreparably impure. The n_{min} sweep says: engage as small as the clustering bears (n_{min}=5), at no purity cost.

6. **The chosen algorithm** (Table 2): a supervised contrastive space (or, best, its contrastive residual child), the density-ratio pool with the derived label-free cut, and the anchor-aware Euclidean statistic. It wins on every dataset/backbone (1.3–1.7× over the best frozen pipeline), and on Galaxy10/LeJEPA the residual-child pipeline reaches $f^\star$=0.013 — *equal to* the transductive GCD ceiling, without ever seeing an unlabelled novel image.

7. **Diffuse shift is the measured boundary of interpretability** (Table 27). A systematic blur of seen-class images is almost perfectly decodable (probe 0.86–1.00) yet the fine-tuned discovery spaces barely register it: the density-ratio tests censor on most cells and mean-shift Mahalanobis — the loser on class novelty — becomes the only reliable detector, while the raw pretrained trunk sees the smear at $f^\star$=0.006–0.010 everywhere. SparKer prefers the one-label (semi-clustered) smear in every engaged comparison: the kernel tests' sharpness for class novelty is a dependence on clusteredness, and a shift with no class structure is detectable but yields nothing an anchor could mean. Practical corollary: pair the discovery pipeline with a mean-shift monitor on the frozen trunk.

8. **The holdout draw dominates the uncertainty** everywhere (ranges span 3–5× within a row; the loop amplifies draw variance ∼5×), so single-draw numbers are never quoted without their spread, and per-draw points are tabulated in full.

arm	draw	frozen SparKer	dist/95th	dist/derived	np/derived
SupCon	h4	0.015	0.027 (MMD)	0.014 (MMD)†	0.014 (MMD)†
	h7	0.021	0.035 (SparKer (anch.))	0.033 (SparKer (anch.))	**0.018 (SparKer)**
	h8	0.018	**0.016 (Eucl. (anch.))**	**0.017 (MMD)**	**0.011 (Eucl. (anch.))**
	h9	0.018	**0.008 (Eucl.)**	**0.011 (Eucl. (anch.))**	**0.009 (SparKer)**
SupCon+SIGReg (λ=5)	h4	0.042	**0.021 (Eucl. (anch.))**	**0.028 (MMD)**	**0.028 (Eucl.)**
	h7	0.034	**0.021 (Eucl. (anch.))**	0.036 (SparKer)†	0.036 (SparKer)†
	h8	0.040	**0.015 (MMD)**	**0.021 (MMD)**	**0.020 (Eucl. (anch.))**
	h9	0.032	**0.016 (Eucl. (anch.))**	0.026 (MMD)†	0.026 (MMD)†
NPLM-dist. (sup.)	h4	0.040	**0.024 (Eucl. (anch.))**	**0.037 (Eucl. (anch.))**	**0.033 (Eucl. (anch.))**
	h7	0.029	**0.022 (Eucl. (anch.))**	0.039 (Eucl. (anch.))	**0.021 (Eucl. (anch.))**
	h8	0.024	**0.016 (SparKer (anch.))**	**0.015 (Eucl. (anch.))**	**0.016 (Eucl. (anch.))**
	h9	0.024	**0.009 (SparKer (anch.))**	**0.020 (Eucl. (anch.))**	**0.020 (Eucl. (anch.))**
NPLM-dist.+classwise SIGReg	h4	0.025	**0.019 (Eucl. (anch.))**	**0.020 (Eucl. (anch.))**	**0.020 (Eucl. (anch.))**
	h7	0.029	**0.018 (Eucl. (anch.))**	**0.027 (SparKer (anch.))**	**0.011 (Eucl. (anch.))**
	h8	0.024	**0.015 (Eucl. (anch.))**	**0.012 (Eucl. (anch.))**	**0.009 (Eucl. (anch.))**
	h9	0.028	**0.016 (Maha.)**	**0.020 (Eucl. (anch.))**	**0.020 (Eucl. (anch.))**

Table 1: Does the discovery loop beat the frozen space? Frozen-space SparKer $f^\star(2\sigma)$ against the best post-discovery test (named in parentheses) under three loop variants: the campaign's distance pool with the 95th-percentile cut, the distance pool with the derived cut, and the density-ratio pool with the derived cut (the paper's construction). Bold = the loop wins on that draw. *Coverage:* CIFAR-10, four parents x draws 4, 7, 8, 9; 48 complete (arm, draw, variant) cells, the loop beats the frozen space in 38; † = the derived cut had declined at the crossing fraction, so that number is the frozen space's and is not counted as a win; '–' = variant not run or still running.

Summary tables

dataset / backbone	best frozen (space / test, median $f^\star$)		best discovery (arm / test)		gain
CIFAR-100 (scratch, 4 draws)	SupCon→res (concat) / SparKer	0.018	SupCon→res-nplm / SparKer[e]	0.008	2.2×
CIFAR-10 (scratch, 4 draws)	SupCon→res-nplm (resid.) / SparKer	0.017	SupCon / Eucl. (anch.)	0.013	1.3×
Galaxy10 / lejepa	SupCon / MMD	0.023	SupCon→res / Eucl. (anch.)	0.013	1.7×
Galaxy10 / visreg	SupCon→res (concat) / MMD	0.026	SupCon→res / Eucl. (anch.)	0.015	1.7×
Galaxy10 / dino	SupCon+SIGReg / MMD	0.029	SupCon / Eucl. (anch.)	0.019	1.5×
transductive ref. (lejepa)	gcd-ft / Maha.	0.013	–	–	sees the class
transductive ref. (visreg)	gcd-ft / Maha.	0.015	–	–	sees the class
transductive ref. (dino)	gcd-ft / MMD	0.028	–	–	sees the class

Table 2: The chosen algorithm, per dataset: best frozen space vs. best discovery pipeline in the same currency. Discovery wins everywhere, always through an anchor-aware test; the transductive GCD rows are the ceiling given the whole novel class unlabelled. *Coverage:* Medians over draws (censored cells imputed at 0.15); CIFAR-10 discovery = np pool + derived cut ($n_{\min}$=10), Galaxy10 discovery = np pool + derived cut ($n_{\min}$=5), both skip-on-refuse; frozen candidates exclude the transductive GCD arms (shown separately). Declined-cut crossings are excluded from the discovery medians; [e] = CIFAR-100 medians cover the engaged draws only (the cut abstains on 33/48 cells at b=0.01).

CIFAR-10: pre-discovery

space	probe	per-ev.	$f^\star(2\sigma)$, median over draws			SparKer range
			Maha.	MMD	SparKer	
SupCon	0.930	0.01	0.057	0.020	0.018	0.015–0.021
→ res (residual)	0.923	0.23	0.023	0.033	0.022	0.012–0.039
→ res (concat)	0.957	0.12	0.024	0.018	0.019	0.016–0.021
→ res-nplm (residual)	0.926	0.22	0.023	0.029	0.017	0.015–0.019
→ res-nplm (concat)	0.946	0.10	0.025	0.020	0.019	0.016–0.021
SupCon+SIGReg (λ=5)	0.912	0.02	$> 0.1^4$	0.031	0.037	0.032–0.042
→ res (residual)	0.902	0.01	$> 0.1^3$	0.044	0.069	0.044–0.091
→ res (concat)	0.949	0.01	$> 0.1^3$	0.035	0.042	0.027–0.060
→ res-nplm (residual)	0.921	0.16	0.044	0.038	0.022	0.017–0.025
→ res-nplm (concat)	0.943	0.06	$> 0.1^2$	0.033	0.030	0.020–0.036
NPLM-dist. (sup.)	0.788	0.17	0.038	0.039	0.027	0.024–0.040
→ res (residual)	0.884	0.03	$> 0.1^2$	0.046	0.046	0.044–0.077
→ res (concat)	0.908	0.03	$> 0.1^2$	0.042	0.043	0.033–0.059
→ res-nplm (residual)	0.841	0.01	$> 0.1^2$	0.044	0.039	0.024–0.049
→ res-nplm (concat)	0.879	0.09	0.047^1	0.045	0.031	0.018–0.043
NPLM-dist.+classwise SIGReg	0.938	0.21	0.032	0.033	0.027	0.024–0.029
→ res (residual)	0.917	0.03	$> 0.1^2$	0.037	0.048	0.039–0.084
→ res (concat)	0.958	0.03	0.034^1	0.035	0.047	0.037–0.078
→ res-nplm (residual)	0.948	0.07	0.034	0.036	0.043	0.028–0.079
→ res-nplm (concat)	0.966	0.17	0.023	0.035	0.037	0.018–0.046
SimCLR	0.808	0.00	$> 0.1^1$	0.027	0.056	0.056–0.056
VISReg/LeJEPA	0.832	0.01	$> 0.1^1$	0.047	0.091	0.091–0.091
NPLM (unsup.)	0.799	0.05	$> 0.1^1$	0.086	$> 0.1^1$	–
SupCon+SIGReg (repulse)	0.871	0.12	0.047	0.039	0.026	0.026–0.026

Table 3: Pre-discovery: minimum injected fraction for a 2σ dataset-level detection, with and without the residual construction. $f^\star$ is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches 2σ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer's range across draws. *Coverage:* CIFAR-10, leakage-free lineage, single holdout; draws 4, 7, 8, 9; 20/24 spaces on every draw; 200 null / 50 signal toys per fraction, N_D=5000; median over draws, superscript = draws where the test never reached 2σ by f=0.1.

space	probe	per-ev.	$f^\star(2\sigma)$, median over draws			SparKer range
			Maha.	MMD	SparKer	
SupCon	0.955	0.03	$> 0.1^2$	0.024	0.023	0.010–0.036
→ res (residual)	0.925	0.05	$> 0.1^3$	0.025	0.028	0.023–0.030
→ res (concat)	0.963	0.05	0.057^1	0.024	0.018	0.017–0.026
→ res-nplm (residual)	0.929	0.10	$> 0.1^3$	0.025	0.027	0.016–0.048
→ res-nplm (concat)	0.963	0.07	$> 0.1^3$	0.023	0.022	0.010–0.030
SupCon+SIGReg (λ=5)	0.947	0.05	0.055	0.022	0.023	0.014–0.026
→ res (residual)	0.906	0.07	$> 0.1^3$	0.028	0.027	0.025–0.032
→ res (concat)	0.959	0.05	0.056	0.020	0.019	0.016–0.026
→ res-nplm (residual)	0.932	0.05	0.055^1	0.022	0.024	0.016–0.027
→ res-nplm (concat)	0.941	0.05	0.059	0.021	0.023	0.014–0.026
NPLM-dist. (sup.)	0.857	0.07	$> 0.1^2$	0.025	0.034	0.017–0.046
→ res (residual)	0.892	0.11	$> 0.1^2$	0.026	0.026	0.025–0.033
→ res (concat)	0.917	0.10	$> 0.1^3$	0.026	0.028	0.026–0.041
→ res-nplm (residual)	0.905	0.08	$> 0.1^2$	0.032	0.026	0.021–0.031
→ res-nplm (concat)	0.924	0.07	$> 0.1^2$	0.028	0.025	0.022–0.029
NPLM-dist.+classwise SIGReg	0.944	0.12	$> 0.1^2$	0.021	0.025	0.015–0.029
→ res (residual)	0.911	0.06	$> 0.1^3$	0.025	0.027	0.024–0.030
→ res (concat)	0.956	0.06	$> 0.1^2$	0.025	0.027	0.023–0.032
→ res-nplm (residual)	0.878	0.04	$> 0.1^3$	0.027	0.027	0.026–0.030
→ res-nplm (concat)	0.955	0.04	$> 0.1^2$	0.022	0.026	0.018–0.030
SimCLR	0.799	0.01	$> 0.1^1$	0.046	0.052	0.052–0.052
VISReg/LeJEPA	0.797	0.03	$> 0.1^1$	0.059	0.047	0.047–0.047
NPLM (unsup.)	0.757	0.01	$> 0.1^1$	0.040	0.031	0.031–0.031
SupCon+SIGReg (repulse)	0.823	0.00	$> 0.1^1$	0.031	0.029	0.029–0.029

Table 4: Pre-discovery: minimum injected fraction for a 2σ dataset-level detection, with and without the residual construction. $f^\star$ is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches 2σ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer's range across draws. *Coverage:* CIFAR-100, leakage-free lineage, single holdout; draws 4, 43, 48, 57; 20/24 spaces on every draw; 200 null / 50 signal toys per fraction, N_D=5000; median over draws, superscript = draws where the test never reached 2σ by f=0.1.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	0.00	0.063	>0.1	0.072	**0.027**	0.036	0.029
→ res (residual)	0.039	0.04	0.074	0.026	0.039	0.043	0.047	**0.026**
→ res-nplm (residual)	0.016	0.08	0.082	**0.029**	0.065	0.045	0.037	0.031
SupCon+SIGReg (λ=5)	0.042	0.01	0.033	**0.021**	>0.1	0.027	0.031	0.039
→ res (residual)	0.091	0.01	0.068	0.043	**0.035**	0.036	0.044	0.045
→ res-nplm (residual)	0.023	0.10	0.027	**0.017**	0.077	0.028	0.029	0.037
NPLM-dist. (sup.)	0.040	0.10	>0.1	**0.024**	0.087	0.046	0.046	0.043
→ res (residual)	0.077	0.01	>0.1	>0.1	0.081	**0.035**	0.043	0.041
→ res-nplm (residual)	0.049	0.01	>0.1	0.047	0.087	0.062	0.083	**0.044**
NPLM-dist.+classwise SIGReg	0.025	0.20	0.034	**0.019**	0.033	0.044	0.028	0.024
→ res (residual)	0.084	0.01	0.027	**0.027**	0.033	0.033	0.029	0.027
→ res-nplm (residual)	0.079	0.03	0.043	**0.022**	0.037	0.036	0.047	0.041

Table 5: Post-discovery $f^\star(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, 95th-pct. cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

CIFAR-10: discovery mode, per draw and loop variant

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	**0.014**†	0.017†	–
→ res (residual)	0.039	0.03	0.042	0.039	0.068	0.039	0.040	**0.032**
→ res-nplm (residual)	0.016	0.01	0.047	0.039	0.049	0.034	0.027	**0.024**
SupCon+SIGReg (λ=5)	0.042	0.00	0.038	0.062	> 0.1	**0.028**	0.047	0.044
→ res (residual)	0.091	0.00	> 0.1	> 0.1	0.078	**0.038**	0.049	0.087
→ res-nplm (residual)	0.023	0.07	0.072	**0.024**	> 0.1	0.039	0.040	0.026
NPLM-dist. (sup.)	0.040	–	0.077	**0.037**	0.072	0.045	0.043	0.043
→ res (residual)	0.077	0.02	> 0.1	> 0.1	> 0.1	**0.038**	0.052	0.060
→ res-nplm (residual)	0.049	–	0.089	0.065	0.089	0.080	**0.049**†	0.074
NPLM-dist.+classwise SIGReg	0.025	–	0.028†	**0.020**	0.074	0.029†	0.028†	0.021
→ res (residual)	0.084	–	> 0.1	0.050	0.083	**0.041**	0.047	0.050
→ res-nplm (residual)	0.079	–	0.049	0.050	0.046	**0.043**	0.067	0.050

Table 6: Post-discovery $f^{\star}(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer $f^{\star}$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\mathrm{seen}} - d_{\mathrm{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	**0.014**†	0.017†	–
→ res (residual)	0.039	0.21	0.033	**0.015**	0.024	0.041	0.033	0.029
→ res-nplm (residual)	0.016	0.14	0.035	**0.020**	0.039	0.041	0.037	0.034
SupCon+SIGReg (λ=5)	0.042	0.13	**0.028**	0.030	0.083	0.034	0.042	0.030
→ res (residual)	0.091	0.15	0.082	**0.030**	0.033	0.042	0.060	0.047
→ res-nplm (residual)	0.023	0.12	0.044	**0.018**	0.074	0.037	0.041	0.021
NPLM-dist. (sup.)	0.040	–	0.087	**0.033**	0.084	0.049	0.073	0.042
→ res (residual)	0.077	0.43	0.056	**0.020**	0.029	0.046	0.046	0.034
→ res-nplm (residual)	0.049	–	0.089	**0.046**	0.089	0.080	0.049†	0.062
NPLM-dist.+classwise SIGReg	0.025	–	0.028†	**0.020**	0.072	0.029†	0.027†	0.036
→ res (residual)	0.084	–	0.079	0.050	0.046	**0.044**	0.045	0.050
→ res-nplm (residual)	0.079	–	0.044	0.050	0.046	**0.043**	0.057	0.050

Table 7: Post-discovery $f^{\star}(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer $f^{\star}$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\mathrm{seen}} - d_{\mathrm{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	>0.1	>0.1	0.043	0.047	0.052	**0.035**
SupCon+SIGReg (λ=5)	0.034	0.03	0.042	**0.021**	>0.1	0.056	0.040	0.043
NPLM-dist. (sup.)	0.029	0.08	0.082	**0.022**	0.038	0.044	0.029	0.023
NPLM-dist.+classwise SIGReg	0.029	0.17	0.029	**0.018**	0.024	0.037	0.028	0.025

Table 8: Post-discovery $f^\star(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	0.071	>0.1	0.036	0.056	0.040	**0.033**
SupCon+SIGReg (λ=5)	0.034	–	0.079	–	0.090	0.079	**0.036**†	–
NPLM-dist. (sup.)	0.029	0.05	>0.1	**0.039**	0.044	0.042	0.042	0.041
NPLM-dist.+classwise SIGReg	0.029	0.10	0.042	0.029	0.035	0.052	0.040	**0.027**

Table 9: Post-discovery $f^\star(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.68	0.022	0.020	0.021	0.052	**0.018**	0.020
→ res (residual)	0.026	0.63	0.017	**0.014**	0.016	0.049	0.018	0.017
→ res-nplm (residual)	0.019	0.61	**0.021**	0.021	0.021	0.065	0.022	0.023
SupCon+SIGReg (λ=5)	0.034	–	0.079	–	0.090	0.072	**0.036**†	–
→ res (residual)	0.089	0.12	0.059	0.030	**0.027**	0.079	0.066	0.046
→ res-nplm (residual)	0.025	0.63	0.039	**0.016**	0.038	0.044	0.028	0.018
NPLM-dist. (sup.)	0.029	0.18	0.044	**0.021**	0.033	0.043	0.041	0.029
→ res (residual)	0.045	–	0.044	0.050	**0.031**	0.076	0.047	0.050
→ res-nplm (residual)	0.049	–	0.092	–	0.089	0.073	**0.072**	–
NPLM-dist.+classwise SIGReg	0.029	0.67	0.043	**0.011**	0.018	0.049	0.033	0.019
→ res (residual)	0.039	0.22	0.036	0.030	**0.023**	0.068	0.029	0.030
→ res-nplm (residual)	0.052	–	0.068	0.050	**0.026**†	0.072	0.048	0.050

Table 10: Post-discovery $f^\star(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.039	**0.016**	0.023	0.018	0.021	0.017
SupCon+SIGReg (λ=5)	0.040	0.01	0.026	0.019	> 0.1	**0.015**	0.026	0.023
NPLM-dist. (sup.)	0.024	0.16	0.037	0.017	0.019	0.019	0.021	**0.016**
NPLM-dist.+classwise SIGReg	0.024	0.18	0.017	**0.015**	0.022	0.022	0.017	0.015

Table 11: Post-discovery $f^\star(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.052	> 0.1	0.038	**0.017**	0.031	0.027
SupCon+SIGReg (λ=5)	0.040	0.01	0.024	0.076	0.062	**0.021**	0.029	0.029
NPLM-dist. (sup.)	0.024	0.59	0.063	**0.015**	0.025	0.023	0.023	0.017
NPLM-dist.+classwise SIGReg	0.024	0.55	0.042	**0.012**	0.039	0.036	0.035	0.021

Table 12: Post-discovery $f^\star(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.45	0.014	**0.011**	0.015	0.019	0.017	0.014
→ res (residual)	0.018	0.42	0.017	0.020	**0.016**	0.019	0.016	0.020
→ res-nplm (residual)	0.018	0.56	0.021	0.015	**0.012**	0.017	0.016	0.015
SupCon+SIGReg (λ=5)	0.040	0.22	0.024	**0.020**	0.024	0.025	0.032	0.035
→ res (residual)	0.044	–	0.046	0.050	0.045	**0.021**†	0.041	0.050
→ res-nplm (residual)	0.021	0.48	0.038	**0.020**	0.025	0.021	0.024	0.036
NPLM-dist. (sup.)	0.024	0.64	0.046	**0.016**	0.041	0.024	0.018	0.017
→ res (residual)	0.046	0.30	0.028	0.030	0.028	**0.022**	0.028	0.030
→ res-nplm (residual)	0.029	0.18	0.027	**0.020**	0.026	0.027	0.027	0.028
NPLM-dist.+classwise SIGReg	0.024	0.73	0.075	**0.009**	0.037	0.044	0.040	0.021
→ res (residual)	0.054	0.40	0.026	**0.011**	0.019	0.019	0.025	0.024
→ res-nplm (residual)	0.033	0.36	0.074	**0.020**	0.027	0.022	0.037	0.020

Table 13: Post-discovery $f^\star(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.02	**0.008**	0.010	0.017	0.018	0.010	0.009
SupCon+SIGReg (λ=5)	0.032	0.01	0.025	**0.016**	0.040	0.040	0.025	0.025
NPLM-dist. (sup.)	0.024	0.14	0.017	0.014	0.015	0.018	0.014	**0.009**
NPLM-dist.+classwise SIGReg	0.028	0.17	0.016	0.022	**0.016**	0.029	0.017	0.017

Table 14: Post-discovery $f^\star(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\mathrm{seen}} - d_{\mathrm{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.013	**0.011**	0.025	0.020	0.014	0.018
SupCon+SIGReg (λ=5)	0.032	–	0.047	> 0.1	0.048	**0.026**†	0.027†	0.050
NPLM-dist. (sup.)	0.024	0.82	0.036	**0.020**	0.024	0.029	0.021	0.027
NPLM-dist.+classwise SIGReg	0.028	0.51	0.026	**0.020**	0.028	0.036	0.026	0.020

Table 15: Post-discovery $f^\star(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\mathrm{seen}} - d_{\mathrm{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.28	0.010	0.013	0.016	0.025	**0.009**	0.010
→ res (residual)	0.012	0.50	0.009	0.010	**0.009**	0.019	0.009	0.010
→ res-nplm (residual)	0.015	0.75	0.014	0.013	0.014	0.016	**0.009**	0.010
SupCon+SIGReg (λ=5)	0.032	–	0.047	0.050	0.045	**0.026**†	0.027†	0.050
→ res (residual)	0.049	0.15	0.035	0.030	**0.027**	0.037	0.028	0.030
→ res-nplm (residual)	0.017	0.59	0.024	**0.010**	0.018	0.028	0.017	0.016
NPLM-dist. (sup.)	0.024	0.84	0.027	**0.020**	0.023	0.029	0.023	0.020
→ res (residual)	0.044	0.16	0.028	0.030	**0.027**	0.034	0.028	0.030
→ res-nplm (residual)	0.024	0.55	0.082	**0.016**	0.027	0.027	0.025	0.026
NPLM-dist.+classwise SIGReg	0.028	0.86	0.044	**0.020**	0.025	0.038	0.025	0.020
→ res (residual)	0.042	–	0.044	0.050	**0.028**†	0.029†	0.036	0.050
→ res-nplm (residual)	0.028	0.20	0.075	0.030	**0.025**	0.044	0.025	0.030

Table 16: Post-discovery $f^\star(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\mathrm{seen}} - d_{\mathrm{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.036	0.07	>0.1	0.023	>0.1	**0.018**	0.028	0.024
→ res (residual)	0.030	–	0.063†	–	>0.1	0.042†	**0.030**†	–
→ res-nplm (residual)	0.048	–	>0.1	–	>0.1	**0.028**†	0.052†	–
SupCon+SIGReg (λ=5)	0.026	–	0.059†	–	0.056†	**0.016**†	0.026†	–
→ res (residual)	0.032	–	0.052†	–	>0.1	0.047†	**0.037**†	–
→ res-nplm (residual)	0.027	–	>0.1	–	>0.1	**0.024**†	0.026†	–
NPLM-dist. (sup.)	0.041	–	>0.1	–	>0.1	**0.019**†	0.048†	–
→ res (residual)	0.033	–	>0.1	–	>0.1	0.047†	**0.034**†	–
→ res-nplm (residual)	0.031	–	>0.1	–	>0.1	0.047†	**0.033**†	–
NPLM-dist.+classwise SIGReg	0.029	0.04	>0.1	0.034	>0.1	0.037	0.033	**0.025**
→ res (residual)	0.030	–	>0.1	–	>0.1	**0.036**†	0.037†	–
→ res-nplm (residual)	0.030	–	>0.1	–	>0.1	0.041†	**0.030**†	–

Table 17: Post-discovery $f^\star(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-100 holdout 4, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

CIFAR-100: discovery mode (np pool, derived cut), per draw

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.023	–	> 0.1	–	0.062†	0.027†	**0.023**†	–
→ res (residual)	0.023	–	0.028†	–	0.047†	0.033†	**0.023**†	–
→ res-nplm (residual)	0.026	–	> 0.1	–	> 0.1	**0.025**†	0.028†	–
SupCon+SIGReg (λ=5)	0.022	–	0.082†	–	0.059†	0.027†	**0.021**†	–
→ res (residual)	0.026	–	0.027†	–	0.040†	0.029†	**0.026**†	–
→ res-nplm (residual)	0.024	–	> 0.1	–	> 0.1	0.028†	**0.023**†	–
NPLM-dist. (sup.)	0.046	–	> 0.1	–	> 0.1	**0.046**†	0.062†	–
→ res (residual)	0.026	–	**0.019**†	–	0.038†	0.025†	0.026†	–
→ res-nplm (residual)	0.026	–	0.084†	–	> 0.1	0.033†	**0.027**†	–
NPLM-dist.+classwise SIGReg	0.027	–	> 0.1	–	> 0.1	0.033†	**0.029**†	–
→ res (residual)	0.024	–	0.027†	–	> 0.1	**0.025**†	0.025†	–
→ res-nplm (residual)	0.027	–	> 0.1	–	> 0.1	0.031†	**0.028**†	–

Table 18: Post-discovery $f^\star(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-100 holdout 43, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.010	–	0.045†	–	> 0.1	0.026†	**0.013**†	–
→ res (residual)	0.028	–	> 0.1	–	> 0.1	**0.016**†	0.029†	–
→ res-nplm (residual)	0.016	0.02	0.015	0.022	0.035	0.013	**0.008**	0.016
SupCon+SIGReg (λ=5)	0.014	0.12	0.024	**0.011**	0.044	0.017	0.016	0.016
→ res (residual)	0.028	–	> 0.1	–	> 0.1	**0.027**†	0.030†	–
→ res-nplm (residual)	0.016	0.08	0.025	**0.014**	0.038	0.016	0.014	0.015
NPLM-dist. (sup.)	0.017	0.35	0.033	0.024	0.029	0.016	0.026	**0.010**
→ res (residual)	0.025	–	0.036†	–	0.031†	**0.017**†	0.023†	–
→ res-nplm (residual)	0.021	–	0.048†	–	0.025†	**0.015**†	0.023†	–
NPLM-dist.+classwise SIGReg	0.015	0.40	0.026	0.014	0.025	0.016	0.015	**0.008**
→ res (residual)	0.025	–	0.077†	–	0.039†	**0.018**†	0.025†	–
→ res-nplm (residual)	0.026	–	> 0.1	–	> 0.1	**0.023**†	0.026†	–

Table 19: Post-discovery $f^\star(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-100 holdout 48, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.024	–	>0.1	–	$0.039^\dagger$	$0.036^\dagger$	$\mathbf{0.027}^\dagger$	–
→ res (residual)	0.029	–	>0.1	–	>0.1	$\mathbf{0.017}^\dagger$	$0.028^\dagger$	–
→ res-nplm (residual)	0.027	–	>0.1	–	>0.1	$0.028^\dagger$	$\mathbf{0.026}^\dagger$	–
SupCon+SIGReg (λ=5)	0.024	–	>0.1	–	$0.059^\dagger$	$0.021^\dagger$	$\mathbf{0.019}^\dagger$	–
→ res (residual)	0.025	–	>0.1	–	>0.1	$\mathbf{0.025}^\dagger$	$0.028^\dagger$	–
→ res-nplm (residual)	0.024	–	$0.052^\dagger$	–	$0.033^\dagger$	$\mathbf{0.017}^\dagger$	$0.021^\dagger$	–
NPLM-dist. (sup.)	0.026	0.07	$0.037^\dagger$	0.023	$0.037^\dagger$	0.025	**0.018**	0.022
→ res (residual)	0.026	–	>0.1	–	>0.1	$\mathbf{0.022}^\dagger$	$0.027^\dagger$	–
→ res-nplm (residual)	0.025	–	$0.076^\dagger$	–	$0.052^\dagger$	$\mathbf{0.024}^\dagger$	$0.026^\dagger$	–
NPLM-dist.+classwise SIGReg	0.023	0.06	0.034	0.035	0.026	0.016	0.016	**0.010**
→ res (residual)	0.029	–	>0.1	–	>0.1	$\mathbf{0.023}^\dagger$	$0.034^\dagger$	–
→ res-nplm (residual)	0.027	–	>0.1	–	>0.1	$\mathbf{0.022}^\dagger$	$0.026^\dagger$	–

Table 20: Post-discovery $f^\star(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer $f^\star$ is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-100 holdout 57, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, N_D=5000; purity = round-1 pool purity at f=0.03; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity '–' = declined at f=0.03; an '(n/m)' row is an arm still running.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.892	0.05	.029	.052	.078	.037	>.1	.015	.048	.031	.022	.023	.015	.073	.066	.035	.042
SupCon+SIGReg (λ=5)	0.762	0.13	.023	>.1	>.1	.033	>.1	.015	.029	.029	.040	.037	.014	>.1	>.1	.028	.043
NPLM-dist. (sup.)	0.787	0.13	>.1	>.1	>.1	.021	>.1	.019	.063	.037	.026	.024	.029	.032	.087	.022	.031
SimCLR	0.829	0.05	.029	.046	>.1	.036	.070	.043	.039	.070	.045	.047	.078	.081	>.1	.095	>.1
SimCLR+SIGReg	0.836	0.06	.048	.065	>.1	.034	>.1	.035	.029	.039	.080	.062	.046	.095	>.1	.046	.093
NPLM-bilinear	0.808	0.08	.078	.042	>.1	.026	>.1	.038	.039	.056	.046	.052	.052	.070	.093	.049	.088
GCD (transductive)	0.899	0.30	.009	.013	.009	.031	.017	.019	.025	.029	.024	.019	.009	.017	.021	.031	.025
GCD+SIGReg (transd.)	0.872	0.48	.015	.019	.019	.043	.025	.014	.035	.040	.042	.029	.013	.027	.026	.030	.028
SupCon → res (residual)	0.910	0.11	.085	.048	>.1	.024	.025	.021	.040	.045	.043	.039	.036	.063	.072	.044	.040
SupCon → res (concat)	0.945	0.12	.084	.041	>.1	.024	.034	.019	.040	.042	.042	.038	.029	.056	.046	.042	.026
SupCon → res-nplm (residual)	0.887	0.08	>.1	.027	>.1	.033	>.1	.016	.041	.073	.045	.032	.029	.072	.089	.040	.078
SupCon → res-nplm (concat)	0.931	0.09	.070	.034	.070	.034	.052	.015	.042	.031	.026	.024	.023	.063	.073	.031	.041
SupCon+SIGReg → res (residual)	0.854	0.06	>.1	.043	>.1	.019	.041	.019	.036	.045	.040	.031	.043	.089	.081	.037	.072
SupCon+SIGReg → res (concat)	0.855	0.09	.031	.063	>.1	.021	.042	.017	.026	.037	.036	.031	.019	.092	.085	.028	.041

Table 21: Galaxy10 / lejepa: pre-discovery $f^\star(2\sigma)$, every (space, draw) point. '>.1' = the test never reached 2σ by f=0.1 on that draw. *Coverage:* Galaxy10 / LeJEPA, 14 spaces × 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, N_D=5000, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

Galaxy10: every point, per backbone

test	d0	d3	d5	d7	d8	median
SupCon — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.019	.044	.038	.023	.027	.027
Eucl. (anch.)	.010	.020	.018	.014	.020	.018
Maha.	.021	.026	.045	.015	.070	.026
MMD	.010	.071	.046	.019	.040	.040
SparKer	.010	.047	.044	.028	.026	.028
SparKer (anch.)	.010	.036	.042	.025	.025	.025
SupCon → res (residual) — engaged from d0: 0.01, d3: 0.01, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	>.1	>.1	>.1	>.1	.150
Eucl. (anch.)	.012	.013	.023	.010	.020	.013
Maha.	.033	.034	.059	.015	.025	.033
MMD	.026	.052	.043	.049	.059	.049
SparKer	.027	.066	.048	.043	.045	.045
SparKer (anch.)	.015	.074	.040	.044	.028	.040
SupCon → res-nplm (residual) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.064	.033	.097	.017	>.1	.064
Eucl. (anch.)	.010	.020	.036	.013	.020	.020
Maha.	.031	.027	>.1	.019	.096	.031
MMD	.021	.045	.069	.047	.037	.045
SparKer	.026	.042	.069	.036	.052	.042
SparKer (anch.)	.017	.027	.042	.041	.028	.028
SupCon+SIGReg (λ=5) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.038	.070	.079	.024	.069	.069
Eucl. (anch.)	.010	.020	.016	.014	.020	.016
Maha.	.018	.047	.072	.023	.056	.047
MMD	.018	.047	.042	.028	.041	.041
SparKer	.017	.062	.084	.019	.043	.043
SparKer (anch.)	.010	.052	.069	.018	.028	.028
SupCon+SIGReg → res (residual) — engaged from d0: 0.01, d3: 0.01, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	.052	>.1	>.1	>.1	.150
Eucl. (anch.)	.015	.018	.025	.011	.020	.018
Maha.	.097	.039	>.1	.014	.029	.039
MMD	.026	.043	.041	.046	.045	.043
SparKer	.026	.088	.085	.041	.049	.049
SparKer (anch.)	.016	.056	.046	.029	.029	.029
NPLM-dist. (sup.) — engaged from d0: never, d3: never, d5: 0.05, d7: 0.1, d8: 0.02; median engaged purity 0.59						
Eucl.	>.1	.070†	>.1	.021†	.029†	.070
Eucl. (anch.)	–	–	–	–	.020	.020
Maha.	>.1	>.1	>.1	.021†	.045	.150
MMD	.016†	.052†	.037	.027†	.027†	.027
SparKer	.029†	.028†	.088†	.023†	.028†	.028
SparKer (anch.)	–	–	–	–	.020	.020

Table 22: Galaxy10 / lejepa: post-discovery $f^\star(2\sigma)$, every (arm, test, draw) point. † = the derived cut had declined at the crossing fraction (the space there is the frozen one); '>.1' = never reached 2σ. *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage:* Galaxy10 / lejepa, discovery mode: density-ratio pool, derived cut, $n_{\min}$=5, skip-on-refuse; head-only loop (2 rounds × 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 172 (test, draw) points.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.906	0.10	>.1	.065	>.1	.042	>.1	.019	.031	.034	.029	.019	.031	.047	.065	.021	.030
SupCon+SIGReg (λ=5)	0.823	0.09	>.1	.065	>.1	.029	>.1	.025	.018	.033	.037	.033	.036	.046	.076	.027	.043
NPLM-dist. (sup.)	0.738	0.13	>.1	.065	>.1	.019	.052	.024	.065	.043	.028	.029	.032	.040	.094	.017	.024
SimCLR	0.847	0.04	.075	.047	>.1	.031	>.1	.049	.044	.066	.052	.049	.072	.093	.090	.082	>.1
SimCLR+SIGReg	0.835	0.05	>.1	.090	>.1	.038	>.1	.040	.026	.044	.070	.046	.069	.091	>.1	.042	.092
NPLM-bilinear	0.672	0.06	>.1	.096	>.1	>.1	>.1	.044	.033	.034	.059	.034	.069	.063	.047	.097	>.1
GCD (transductive)	0.886	0.40	.015	.009	.012	.041	.018	.024	.021	.026	.029	.028	.010	.015	.017	.027	.019
GCD+SIGReg (transd.)	0.852	0.48	.013	.017	.026	.031	.026	.017	.024	.039	.033	.027	.022	.022	.032	.029	.019
SupCon $\rightarrow$ res (residual)	0.930	0.06	>.1	.037	>.1	.043	.037	.016	.025	.035	.036	.036	.026	.087	.082	.047	.043
SupCon $\rightarrow$ res (concat)	0.947	0.09	>.1	.040	.084	.031	.042	.019	.026	.034	.026	.025	.025	.045	.063	.028	.029
SupCon $\rightarrow$ res-nplm (residual)	0.890	0.11	>.1	.052	>.1	.024	.073	.019	.036	.033	.047	.027	.031	.065	.081	.022	.029
SupCon $\rightarrow$ res-nplm (concat)	0.923	0.12	>.1	.048	>.1	.033	.075	.018	.035	.033	.029	.024	.029	.049	.071	.023	.029
SupCon+SIGReg $\rightarrow$ res (residual)	0.867	0.05	>.1	.056	>.1	.017	.052	.025	.024	.039	.045	.038	.038	.086	.080	.043	.062
SupCon+SIGReg $\rightarrow$ res (concat)	0.888	0.09	>.1	.042	>.1	.019	.036	.021	.021	.033	.040	.029	.029	.043	.052	.030	.037

Table 23: Galaxy10 / visreg: pre-discovery $f^\star(2\sigma)$, every (space, draw) point. '>.1' = the test never reached 2σ by f=0.1 on that draw. *Coverage:* Galaxy10 / VISREG, 14 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, N_D=5000, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
SupCon — engaged from d0: 0.01, d3: 0.02, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.070	.038	.063	.024	.028	.038
Eucl. (anch.)	.015	.020	.021	.019	.020	.020
Maha.	>.1	.024	.056	.021	.042	.042
MMD	.017	.037	.040	.029	.027	.029
SparKer	.025	.056	.063	.024	.033	.033
SparKer (anch.)	.022	.042	.044	.021	.021	.022
SupCon → res (residual) — engaged from d0: 0.01, d3: 0.02, d5: 0.02, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	>.1	>.1	>.1	.150
Eucl. (anch.)	.015	.020	.024	.014	.014	.015
Maha.	.045	.042	.066	.019	.033	.042
MMD	.031	.026	.029	.026	.045	.029
SparKer	.025	.054	.084	.038	.039	.039
SparKer (anch.)	.013	.041	.077	.024	.029	.029
SupCon → res-nplm (residual) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	.034	.037	.052	.022	.025	.034
Eucl. (anch.)	.015	.021	.025	.013	.019	.019
Maha.	.056	.024	.049	.015	.028	.028
MMD	.024	.040	.039	.042	.038	.039
SparKer	.026	.048	.047	.024	.024	.026
SparKer (anch.)	.017	.025	.044	.020	.027	.025
SupCon+SIGReg (λ=5) — engaged from d0: 0.01, d3: 0.01, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.077	.048	>.1	.016	>.1	.077
Eucl. (anch.)	.017	.022	.016	.013	.020	.017
Maha.	>.1	.026	.097	.023	.059	.059
MMD	.019	.036	.039	.041	.034	.036
SparKer	.034	.030	.062	.026	.045	.034
SparKer (anch.)	.018	.029	.040	.017	.025	.025
SupCon+SIGReg → res (residual) — engaged from d0: 0.02, d3: 0.01, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	>.1	>.1	>.1	>.1	.150
Eucl. (anch.)	.020	.018	.031	.010	.020	.020
Maha.	.063	.034	.072	.015	.027	.034
MMD	.028	.033	.043	.038	.029	.033
SparKer	.031	.052	.078	.044	.047	.047
SparKer (anch.)	.021	.034	.042	.037	.026	.034
NPLM-dist. (sup.) — engaged from d0: 0.02, d3: 0.05, d5: 0.1, d7: 0.03, d8: 0.1; median engaged purity 0.43						
Eucl.	.082†	.041	>.1	.015†	.026†	.041
Eucl. (anch.)	–	–	–	.030	–	.030
Maha.	>.1	.039	>.1	.016†	.044†	.044
MMD	.018	.044	.044†	.025	.037†	.037
SparKer	.034†	.032	.087	.017†	.027†	.032
SparKer (anch.)	–	–	–	.030	–	.030

Table 24: Galaxy10 / visreg: post-discovery $f^\star(2\sigma)$, every (arm, test, draw) point. † = the derived cut had declined at the crossing fraction (the space there is the frozen one); '>.1' = never reached 2σ. *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage:* Galaxy10 / visreg, discovery mode: density-ratio pool, derived cut, $n_{\min}$=5, skip-on-refuse; head-only loop (2 rounds × 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 172 (test, draw) points.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.879	0.06	>.1	>.1	.063	.076	>.1	.031	.029	.025	.042	.037	.078	>.1	>.1	.049	.083
SupCon+SIGReg (λ=5)	0.804	0.04	>.1	.072	.069	>.1	>.1	.021	.025	.031	.047	.029	>.1	>.1	>.1	.079	>.1
NPLM-dist. (sup.)	0.775	0.10	>.1	>.1	>.1	.059	>.1	.037	.023	.029	.038	.037	.054	.086	.070	.038	.048
SimCLR	0.790	0.05	>.1	.066	>.1	>.1	>.1	.029	.039	.056	.067	.052	.070	.097	>.1	.066	>.1
SimCLR+SIGReg	0.801	0.04	>.1	.069	>.1	.076	>.1	.027	.029	.041	.088	.059	.069	.084	>.1	.059	.091
NPLM-bilinear	0.684	0.04	>.1	>.1	.082	>.1	>.1	.097	.042	.041	.093	>.1	.081	.089	.079	.083	.089
GCD (transductive)	0.821	0.07	>.1	.052	.037	.076	>.1	.019	.025	.031	.040	.028	.059	.087	.085	.069	.092
GCD+SIGReg (transd.)	0.831	0.12	.084	.066	.056	.073	>.1	.024	.026	.034	.039	.034	.052	>.1	.096	.066	.098
SupCon $\rightarrow$ res (residual)	0.891	0.05	>.1	.080	.062	.097	>.1	.023	.029	.047	.056	.033	.090	.098	.097	.081	.097
SupCon $\rightarrow$ res (concat)	0.903	0.06	>.1	>.1	.063	.074	>.1	.025	.025	.029	.048	.034	.074	>.1	.098	.073	.092
SupCon $\rightarrow$ res-nplm (residual)	0.862	0.07	>.1	>.1	.056	.087	>.1	.024	.024	.036	.039	.032	.077	>.1	>.1	.047	.092
SupCon $\rightarrow$ res-nplm (concat)	0.890	0.06	>.1	>.1	.063	.075	>.1	.029	.026	.027	.042	.033	.048	.093	>.1	.047	.092
SupCon+SIGReg $\rightarrow$ res (residual)	0.852	0.06	>.1	.073	.092	>.1	>.1	.025	.025	.044	.049	.032	.088	>.1	.096	.082	>.1
SupCon+SIGReg $\rightarrow$ res (concat)	0.869	0.05	>.1	.079	.072	.096	>.1	.021	.025	.036	.047	.033	.089	>.1	>.1	.075	>.1

Table 25: Galaxy10 / dino: pre-discovery $f^\star(2\sigma)$, every (space, draw) point. '>.1' = the test never reached 2σ by f=0.1 on that draw. *Coverage:* Galaxy10 / DINO, 14 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, N_D=5000, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
SupCon — engaged from d0: 0.01, d3: 0.02, d5: 0.03, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	.090	.022	>.1	>.1	.150
Eucl. (anch.)	.019	.021	.030	.014	.018	.019
Maha.	>.1	>.1	.066	>.1	>.1	.150
MMD	.029	.039	.036	.059	.044	.039
SparKer	.062	.079	>.1	.048	.079	.079
SparKer (anch.)	.044	.044	.089	.035	.042	.044
SupCon → res (residual) — engaged from d0: 0.02, d3: 0.01, d5: 0.01, d7: 0.02, d8: 0.02; median engaged purity 1.00						
Eucl.	>.1	.078	>.1	>.1	>.1	.150
Eucl. (anch.)	.020	.021	.025	.020	.031	.021
Maha.	>.1	.070	.069	.085	>.1	.085
MMD	.052	.028†	.066	.087	.066	.066
SparKer	.040	.083	.096	.078	>.1	.083
SparKer (anch.)	.039	.036	.049	.044	.044	.044
SupCon → res-nplm (residual) — engaged from d0: 0.01, d3: 0.01, d5: 0.02, d7: 0.02, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	.072	.036	>.1	>.1	.150
Eucl. (anch.)	.016	.019	.038	.020	.019	.019
Maha.	>.1	>.1	.041	.068	>.1	.150
MMD	.027	.039	.040	.050	.036	.039
SparKer	.046	.084	.092	.069	.097	.084
SparKer (anch.)	.045	.037	.052	.044	.043	.044
SupCon+SIGReg (λ=5) — engaged from d0: 0.02, d3: 0.01, d5: 0.01, d7: never, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	.062	>.1	>.1	.150
Eucl. (anch.)	.020	.019	.022	–	.031	.021
Maha.	>.1	.075	.078	>.1	>.1	.150
MMD	.027	.029	.039	.046†	.046	.039
SparKer	.048	.077	.084	.075†	>.1	.077
SparKer (anch.)	.044	.046	.047	–	.052	.047
SupCon+SIGReg → res (residual) — engaged from d0: 0.01, d3: 0.01, d5: 0.01, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	.096†	>.1	>.1	.150
Eucl. (anch.)	.014	.026	.026	–	.034	.026
Maha.	.097	.086	>.1	>.1	>.1	.150
MMD	.066	.028†	.047	.052†	.048	.048
SparKer	.059	>.1	.074†	.084†	>.1	.084
SparKer (anch.)	.047	.074	.043	–	.046	.047
NPLM-dist. (sup.) — engaged from d0: never, d3: never, d5: never, d7: never, d8: 0.03; median engaged purity 0.28						
Eucl.	.071†	>.1	>.1	.018†	>.1	.150
Eucl. (anch.)	–	–	–	–	.030	.030
Maha.	>.1	>.1	>.1	.072†	>.1	.150
MMD	.035†	.024†	.029†	.037†	.052	.035
SparKer	.045†	.078†	.056†	.045†	.094	.056
SparKer (anch.)	–	–	–	–	.047	.047

Table 26: Galaxy10 / dino: post-discovery $f^\star(2\sigma)$, every (arm, test, draw) point. † = the derived cut had declined at the crossing fraction (the space there is the frozen one); '>.1' = never reached 2σ. *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage:* Galaxy10 / dino, discovery mode: density-ratio pool, derived cut, $n_{\min}$=5, skip-on-refuse; head-only loop (2 rounds × 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 168 (test, draw) points.

space	one label (class 2)					all seen labels				
	out. AUC	probe	Maha.	MMD	SparKer	out. AUC	probe	Maha.	MMD	SparKer
lejepa										
pretrained trunk	0.86	1.00	.007	.042	.022	0.89	1.00	.006	.023	.042
SupCon	0.69	0.99	.079	>.1	.036	0.51	0.95	.016	.083	>.1
SupCon+SIGReg (λ=5)	0.57	0.99	.028	>.1	.035	0.57	0.94	.031	>.1	.082
NPLM-dist. (sup.)	0.48	0.88	.013	>.1	>.1	0.58	0.86	.027	>.1	>.1
visreg										
pretrained trunk	0.94	1.00	.006	.047	.008	0.94	1.00	.006	.024	.016
SupCon	0.52	1.00	.022	>.1	.023	0.59	1.00	.022	.083	.056
SupCon+SIGReg (λ=5)	0.84	1.00	.007	>.1	.009	0.69	1.00	.013	>.1	.037
NPLM-dist. (sup.)	0.35	0.77	.093	>.1	>.1	0.62	0.64	.068	.086	.088
dino										
pretrained trunk	0.77	1.00	.010	.049	.020	0.82	1.00	.008	.027	.080
SupCon	0.61	1.00	.013	>.1	.023	0.63	1.00	.010	.076	.060
SupCon+SIGReg (λ=5)	0.68	1.00	.024	>.1	.024	0.63	0.99	.022	.059	.067
NPLM-dist. (sup.)	0.76	0.81	.031	>.1	.043	0.63	0.69	.063	>.1	.086

Table 27: Diffuse-shift detection: a systematic smear of seen-class images instead of a novel class. The smear stays almost perfectly *decodable* (probe 0.86–1.00) but the fine-tuned discovery spaces barely see it as outlying; mean-shift Mahalanobis becomes the only reliable test there, while the raw pretrained trunk detects it at $f^\star$=0.006–0.010 everywhere — the systematics monitor. SparKer prefers the semi-clustered one-label smear in every engaged comparison: the kernel tests' class-novelty sharpness is a dependence on clusteredness. *Coverage:* Galaxy10, 24 (backbone, space, mode) cells; smear = Gaussian blur σ=2 px applied identically to $\leq$1000 seen-class test images (source rows removed from the background pool); exp-146 toy battery. *out. AUC* = min-anchor-distance ROC of smeared vs. background; *probe* = smeared-vs-clean linear AUC.

Galaxy10: diffuse-shift boundary

space	novel class 2 (held out)			smeared class 2 (seen, blur σ=2)			cost
	Maha.	MMD	SparKer	Maha.	MMD	SparKer	
lejepa							
pretrained trunk	>.1	.031	.079	.007	.042	.022	0.2×
SupCon	.029	.015	.015	.079	>.1	.036	2.4×
SupCon+SIGReg (λ=5)	.023	.015	.014	.028	>.1	.035	1.9×
NPLM-dist. (sup.)	>.1	.019	.029	.013	>.1	>.1	0.7×
visreg							
pretrained trunk	>.1	.034	.073	.006	.047	.008	0.2×
SupCon	>.1	.019	.031	.022	>.1	.023	1.1×
SupCon+SIGReg (λ=5)	>.1	.025	.036	.007	>.1	.009	0.3×
NPLM-dist. (sup.)	>.1	.024	.032	.093	>.1	>.1	3.9×
dino							
pretrained trunk	>.1	.040	.049	.010	.049	.020	0.3×
SupCon	>.1	.031	.078	.013	>.1	.023	0.4×
SupCon+SIGReg (λ=5)	>.1	.021	>.1	.024	>.1	.024	1.1×
NPLM-dist. (sup.)	>.1	.037	.054	.031	>.1	.043	0.8×

Table 28: Same label, two anomaly types: class novelty vs. diffuse smear. The reduction is per test family, not blanket: the clustering-sensitive tests lose heavily on the smear (MMD censors on 8 of 9 fine-tuned rows; SparKer costs 1.5–2.4× where both engage), while the mean-shift Mahalanobis often detects the smear *cheaper* than novelty (costs down to 0.3×) — but a mean-shift detection carries no class structure for an anchor to mean. The raw pretrained trunk sees the smear 3–5× better than it sees the same class as novelty (cost 0.2–0.3×): the systematics monitor. *Coverage:* Galaxy10, 12 (backbone, space) pairs, both anomalies built from CLASS 2: left = class 2 held out of training and injected as a novel class (exp-150 draw d0); right = class 2 seen in training, a subset smeared and injected (exp 152). The pretrained-trunk row uses the identical space on both sides; the fine-tuned rows differ in which class the fine-tune excluded (2 vs. the archived 9). cost = ratio of best-test $f^\star$, smeared over novel.